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1. JS :-

⇒ Java Script is a programming language used to make webpages interactive and dynamic.

⇒ It is a high level programming language. It is mainly used to front-end development but it can also be used in back-end development. with node.js.

⇒ Java Script work with HTML node.js to create interactive websites.

⇒ For example JavaScript is used for button clicks form validation, changing content and showing or hiding elements.

⇒ It is a popular programming language used to in modern web development.

2. Diff var, let and const :

* Var, let and const are keyword used to declare variable in JS.

* And Es+6 feature introduced let and const.

var :-

* Function - scope
* can be redeclared
* can be reassigned.

let :

* Block - Scope
* cannot be redeclared in same scope.
* Reassigned.

Const :

* Block - Scope
* cannot Redeclared and Reassigned.

3) Different Data types JS :-

Data types specify what kind of value a variable can store in JS.

i) primitive - string, Num, boolean, Null, undefined.
ii) Non - primitive - Array, obj, Arr of obj

i) primitive :-

- primitive Data types store a single, simple value, they are Immutable.

ii) Non - primitive :-

Non - primitive Data types are used to store collections of data or complex data.

4) Functions in JS :

A function is a reusable block of code that performs a specific task, we can call a function whenever we need to executed that task.

we call a function through argument and that argument we pass value that value goes to parameter.

i) Function Declaration.

It is a traditional method, It can be called before its declaration.

ii) Function Expression :

A function is stored Inside a variable.

iii) Arrow function (=>)

A shorter way to write function

It is a synatic sugar.

An arrow function is a shorter way to write a function using the => symbol.

Paragraph 2: Each form element has attributes such as id, name, type, placeholder, and required. JavaScript can select these

5) Difference between == and === JS;

=> Both == and === comparison operators used to compare two values.

=> == is make sign it's only compare value not Datatypes. (Loose equality)

=> === Strong sign it's compare both values and Datatype. (Strict equality)

6) React JS ?

* React JS is JS Library developed by meta for building user interface especially single page application.

1) Component - Based Architecture.
2) Virtual DOM.
3) Reusable code -> props
4) Fast UI Development.

7) component :-

=> A component is a small, independent, reusable part of a React application. Each component represents a part of user interface.

Ex:- Navbar, footer, Login, Register.

=> we can use same component in different pages without writing same code.

=> There are JS function that return JSX.

=> Each component can manage it's own logic and UI.

8) Virtual DOM :

⇒ Virtual DOM is lightweight is representation of Real DOM. React uses it to identify changes in user interface and update necessary parts of actual DOM.

⇒ Diffing - React compares the previous and new Render o/p to identify what was changed.

⇒ Reconciliation - React applies the necessary update to the real DOM instead of unnecessily rebuilding the page.

9) What is JSX in react ?

⇒ JSX stands for JavaScript XML. It is a syntax Extension for JS that allows us to write HTML-like code inside JS.

⇒ HTML-like syntax : JSX allows us write UI elements using HTML like syntax.

⇒ Compilation : Browsers do not directly understand JSX. tools such as React build toolchain

⇒ Easily to understand : JSX makes react code easier to read and maintained.

10. What is diff between props and state in react ? Explain how data is passed between components and how state updates the UI.

props

Props are used to pass data from one component to another component. props means Properties. props are passed from parent Component to child component. Props are read only - we cannot change props directly.

ments
different form elements to collect information from users. The <form> element groups related input fields, while
nd <textarea> allow users to enter different types of information.
ement has attributes such as id, name, type, placeholder, and required. JavaScript can select these



Child component Example.

```
function Stu(props){
    return <h1> Hello {props.name} </h1>
}

function App(){
    return <Stu name = "Nathi" />.
}
```

o/p :- Hello Nathi.

Parent compon

State.

State is used to store and manage data inside a component. State is created using useState. that state can be changed using its update function. When state changes React updates the UI.

We use state of the useState hook to create state.

So props are used to pass data and state is used to manage data.